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From Logs to Spectra: A Data Driven Framework for Synthetic NMR T_2 Distribution in Carbonate Reservoir

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Abstract

Accurate quantification of porosity distribution, permeability, and water saturation is fundamental for detailed petrophysical characterization and reservoir evaluation. While basic well log data enable estimation of total and effective porosity, reliable determination of porosity types, permeability, and irreducible water saturation often requires supplemental laboratory core analyses or advanced logging techniques. Among these, nuclear magnetic resonance (NMR) logging has emerged as a transformative technology, providing detailed pore system characterization at each depth point based on the full T2 distribution rather than a single scalar value. This enables evaluation of key reservoir properties including clay bound fluid, capillary bound fluid, free fluid volume, and permeability. However, the deployment of NMR tools is frequently limited by high operational costs, challenging borehole environments, suboptimal signal-to-noise ratios, and magnetic interference from drilling fluids, resulting in significant data gaps across many wells. To overcome these limitations, this study leverages machine learning (ML) techniques to predict NMR-derived reservoir properties from conventional well log inputs, thereby improving data availability without the need for extensive NMR tool runs. We conducted a systematic evaluation of five supervised ML algorithms —Random Forest, AdaBoost, Gradient Boosting, XGBoost, and CatBoost—assessing performance using R^2, mean squared error (MSE), and root mean squared error (RMSE) metrics. Among these, the CatBoost model demonstrated superior predictive accuracy and robustness, particularly during blind well validation, underscoring its suitability for petrophysical applications. Building upon these predictions, we developed a novel workflow for reconstructing the complete NMR T_2 relaxation time distribution using statistical distribution functions derived from predicted parameters. This methodology produced synthetic T_2 spectra exhibiting strong concordance with field-acquired NMR log data, thereby validating our approach for practical reservoir assessments. The proposed framework facilitates cost-effective, detailed reservoir characterization in data-limited environments, empowering operators to enhance reservoir management strategies and recovery forecasts even in the absence of direct NMR measurements. This work thus introduces a robust data-driven solution for filling NMR data gaps and advancing digital petrophysical workflows.

Keywords: Reservoir Petrophysics, Nuclear Magnetic Resonance Logging, Machine Learning, T_2 Distribution, Permeability Estimation

Introduction

Accurate reservoir characterization demands a detailed understanding of essential transport and storage properties, including pore size distribution, permeability, and capillary behavior. The quantification of these properties is crucial in disciplines such as hydrocarbon exploration and production (Tiab and Donaldson, 2016; Su et al. 2020) groundwater studies (Kang et al. 2017), geothermal reservoirs (Siler et al. 2019), carbon capture and storage (Zhang et al. 2017), and underground hydrogen storage projects. Conventional well log data primarily provide insights into overall and shale-corrected effective porosity, yet do not reveal detailed distributions of displaceable versus non-displaceable fluid within pore networks or pore size diversity. Critically, fluid flow parameters such as permeability are difficult to determine directly from standard well logs, which are optimized for porosity assessment. Their estimation commonly relies on empirical correlations (Kozney 1927; Carman 1937; Archie 1942), time-consuming laboratory measurements or well pressure tests (Horner 1951; Jensen and Mayson 1985). Similarly, capillary-related volumes—most importantly irreducible water saturation (Swirr), denoting water entrapped in small pores—require thorough core measurements and interpretation (El Sharawy and Gaafar 2019). In recent decades, nuclear magnetic resonance (NMR) logging has gained substantial popularity for quantifying different reservoir porosity volumes, permeability, and capillary effects (Mondal and Singh, 2023b). NMR tools, unlike traditional mineralogy-dependent logs, offer direct insight into pore volume distributions, fluid compositions, and enable precise assessments of hydrodynamic properties (Liu et al. 2016). Despite their utility, widespread adoption of NMR is hampered by high implementation costs, ineffectiveness in cased-hole applications, and an absence of NMR data in wells drilled prior to the 1990s (Masroor et al. 2022; Tamoto et al. 2023). As a result, there is a growing research focus on generating synthetic NMR-equivalent outputs from existing conventional well logs. Machine learning models are emerging as a highly effective toolkit for predicting such critical reservoir properties (Rezaee 2022). Masroor et al. (2022) demonstrated improved performance of nonlinear machine and deep learning algorithms for estimating NMR-derived permeability from traditional logs. Razee (2022) emphasizes AdaBoost's reliability for predicting NMR outputs in clastic rocks. Tariq et al. (2023) employed Random Forest to model NMR-based effective porosity in carbonates with practical accuracy, while Tamoto et al. (2023) compared multilayer perceptron, AdaBoost, XGBoost, and CatBoost for synthetic NMR porosity log generation. Yet, there remains a literature gap regarding the end-to-end modelling of the full T_2 spectral distribution directly from NMR outputs, as previous studies typically fit Gaussian curves directly to T_2 spectra (Genty et al., 2007; Ge et al., 2014). In contrast, we modeled T_2 spectra using ML outputs combined with statistical methods, allowing non-linear relationships where necessary instead of assuming a fixed distribution. In this study, we analyze data from one carbonate formation, characterized by diverse pore networks and diagenetic textures (Mondal and Singh, 2021; 2022; 2023a; 2023b; 2024). Most wells in this case lack advanced log data, having been drilled before 1990, with only limited NMR coverage. Our approach deploys ensemble machine learning models—Random Forest, AdaBoost, Gradient Boosting, XGBoost, and CatBoost—to reconstruct standard NMR outputs, confirming CatBoost's superior predictive accuracy. Our python-based workflow additionally models the entire T_2 spectrum using distribution functions. This innovative methodology not only enhances the estimation of porosity distribution but also extends NMR utility to wells lacking direct NMR data, marking a pioneering advance in digital Petrophysics.

Fundamentals of Nuclear Magnetic Resonance (NMR)

Nuclear magnetic resonance (NMR) logging is an essential technique used in several fields such as oil and gas exploration, groundwater analysis, carbon capture and storage, and hydrogen storage (Nandi et al., 2020;

Mondal et al., 2021; Roshdy et al., 2025). It quantifies the signal produced by the magnetization of hydrogen nuclei (equivalent to protons). The signal strength is governed by the total count of hydrogen nuclei within the fluid, as modelled from NMR T_2 decay spectra (Fig. 1a) obtained from CPMG measurement (Coates et al. 1999), which is directly associated with the porosity of rock or rock-fluid interaction (Ge et al., 2016). T_2 distributions are obtained by inverting NMR decay spectra, typically within the industry-standard range of 0.3 to 3000 ms (Fig. 1b). The NMR T_2 spectra of saturating fluids, like water/hydrocarbon, provides insights into properties such as porosity, pore size distribution, permeability etc. (Mondal and Singh, 2023a; 2023b). Hydrogen nuclei located near the surface of shale or clay-bound water (CBW), as well as those in fluid within relatively small pores (small size pores with capillary water: BVI), demonstrate short T_2 relaxation times, while fluid occupying larger pores exhibits longer T_2 times. Typically, designated cut-off values of T_2 serve to differentiate between CBW, irreducible capillary pore volume (Bulk Volume Irreducible or BVI), and movable or producible fluid volume (Free Fluid Index, FFV) (Prammer et al. 1996; Coates et al. 1999). The cut-off is determined either experimentally using NMR curves of core samples at different saturation levels or assumed based on lithology. For instance, typically set at 3 ms for clay-bound porosity and either ~33 ms or ~100 ms for capillary-bound pores, depending on whether the rock is clastic or carbonate, respectively (Fig. 1b). Standard practice involves classifying T_2 distribution into different pore volumes to get comprehensions into diverse pore systems within rocks using the following equations:

$$\mathrm{CBW} = \sum_{T_2=0.3}^{3} V(T_2) \tag{1}$$

$$\mathrm{BVI} = \sum_{T_2=3}^{T_{2c}} V(T_2) \tag{2}$$

$$\mathrm{FFV} = \sum_{T_{2c}}^{3000} V(T_2) \tag{3}$$

$$\mathrm{Swirr} = \frac{\mathrm{BVI}}{\mathrm{BVI}+\mathrm{FFV}}. \tag{4}$$

$$\text{Total Porosity }(\mathrm{MRP}) = \mathrm{CBW}+\mathrm{BVI}+\mathrm{FFV} = \mathrm{BFV}+\mathrm{FFV} \tag{5}$$

Where, T_2 = Transverse relaxation time; V=Volume of the pore, and Swirr = irreducible water saturation; T_2 cut-off values (T_{2c}) are decided considering nature of formation; BFV (Bound fluid volume: Immovable fluid fraction) includes both CBW and BVI.

The area under the T_2 distribution curve equals the total NMR porosity (MRP). The logarithmic mean of the T_2 distribution (T_{2LM}) is determined for each sample based on the following equation:

$$T_{2LM} = \exp\left(\frac{\sum \log T_{2i} \cdot \varphi_i}{\sum \varphi_i}\right) \tag{6}$$

Where, T_{2i} and φ_i are the T_2 relaxation time and porosity of the i^{th} porosity component in the distribution.

Permeability estimation from NMR T_2 data often employs models such as the Coates (Coates et al., 1999) and SDR (Schlumberger-Doll-Research) models (Steen et al., 2012). The Coates model relies on a T_2 cut-off value to distinguish between Free Fluid Index (FFV) and Bulk Volume Irreducible (BVI), while the SDR model utilizes the logarithmic mean of the T_2 distribution (T_{2LM}).

$$K_c = \left[\left(\frac{\mathrm{MRP}}{C_1}\right)^{m_1} \cdot \left(\frac{\mathrm{FFV}}{\mathrm{BVI}}\right)^{n_1}\right] \tag{7}$$

$$K_{SDR} = \left[C_2(\mathrm{MRP})^{m_2} \cdot T_{2LM}^{n_2}\right] \tag{8}$$

m_1, n_1, C_1, m_2, n_2, and C_2 are the statistical model parameters, whose values can be derived from lab NMR experimental data sets of core samples. In the absence of core samples, empirical values were assigned accordingly.

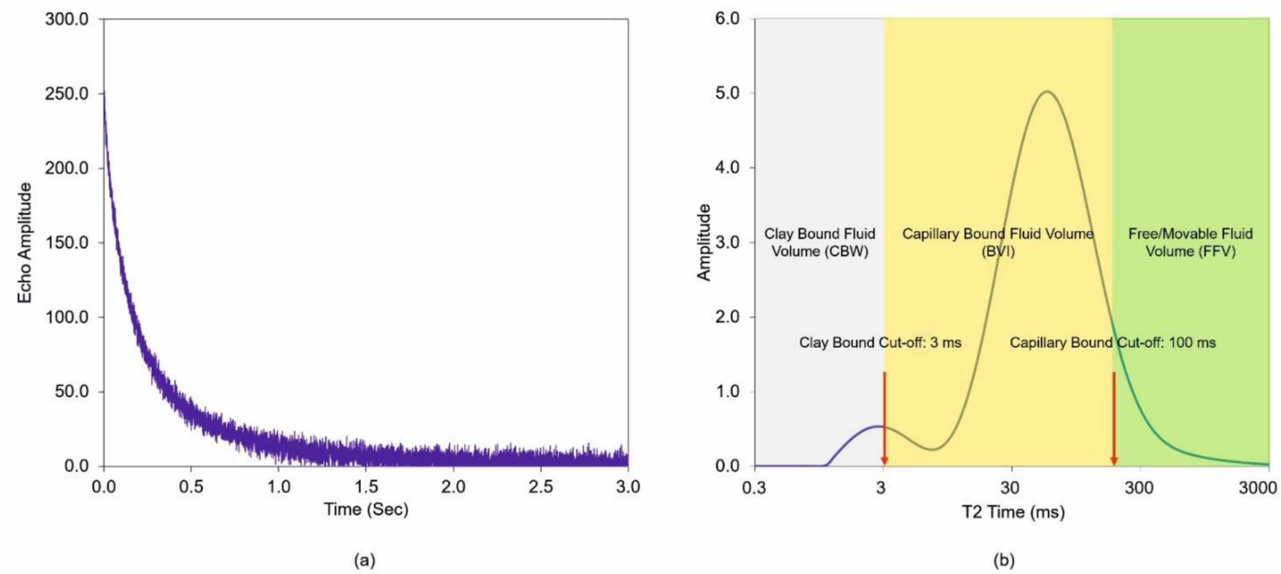

Figure 1—(a) Representation of NMR multi-exponential decay curves obtained from CPMG measurements, (b) Inverted T_2 distribution illustrating the characterization of clay-bound water (CBW), capillary-bound fluid (BVI), and free fluid (FFV) in carbonate, with standard default cutoff values of 3 msec and 100 msec highlighted.

Data Preparation and Feature Selection

Our workflow (Fig. 2) applies ensemble machine learning algorithms to model key NMR outputs using standard well log data from the carbonate formation. The dataset consists of ~9000 depth-matched data points from three wells with acquired NMR logs. The three wells examined are located within 500 to 800 meters of each other and lie within the same structural feature and depositional setting, characterized as a mid-ramp carbonate. Lithofacies and log responses are broadly consistent across the wells, with only minor facies variations observed. All logs were quality-checked and corrected for depth mismatches, borehole anomalies, and log spikes. Hydrocarbon correction was applied to resistivity and porosity logs to improve petrophysical accuracy. Precision depth alignment between NMR and conventional logs was performed, leveraging gamma ray (GR) data across both datasets. Input features included caliper (CALI), gamma ray (GR), deep resistivity (RT), neutron porosity (NPHI), bulk density (RHOB), density correction (DRHO), compressional sonic (DT), shale volume (V_{sh}), difference between neutron and density porosity (DPHI), and shale-corrected total porosity, commonly known as effective porosity (PHIE). Shale volume was averaged from GR (V_{shGR}) and density-neutron (V_{shND}) methods (Serra,1986). DPHI used standard limestone as matrix and water as fluid assumptions, while PHIE accounted for shale effects in averaged density, neutron, and sonic based total porosities. Targets included clay bound water (CBW), bulk volume irreducible (BVI), free fluid volume (FFV), Coates and SDR permeability (K_c, K_{SDR}), and log mean T_2 (T_{2LM}). Correlation analysis identified weak input-NMR associations for CALI and DRHO, resulting in their exclusion. After combining two wells' data and randomization, 80% was used for model training and 20% for testing & validation. The remaining one well was reserved for blind validation to ensure robust performance assessment.

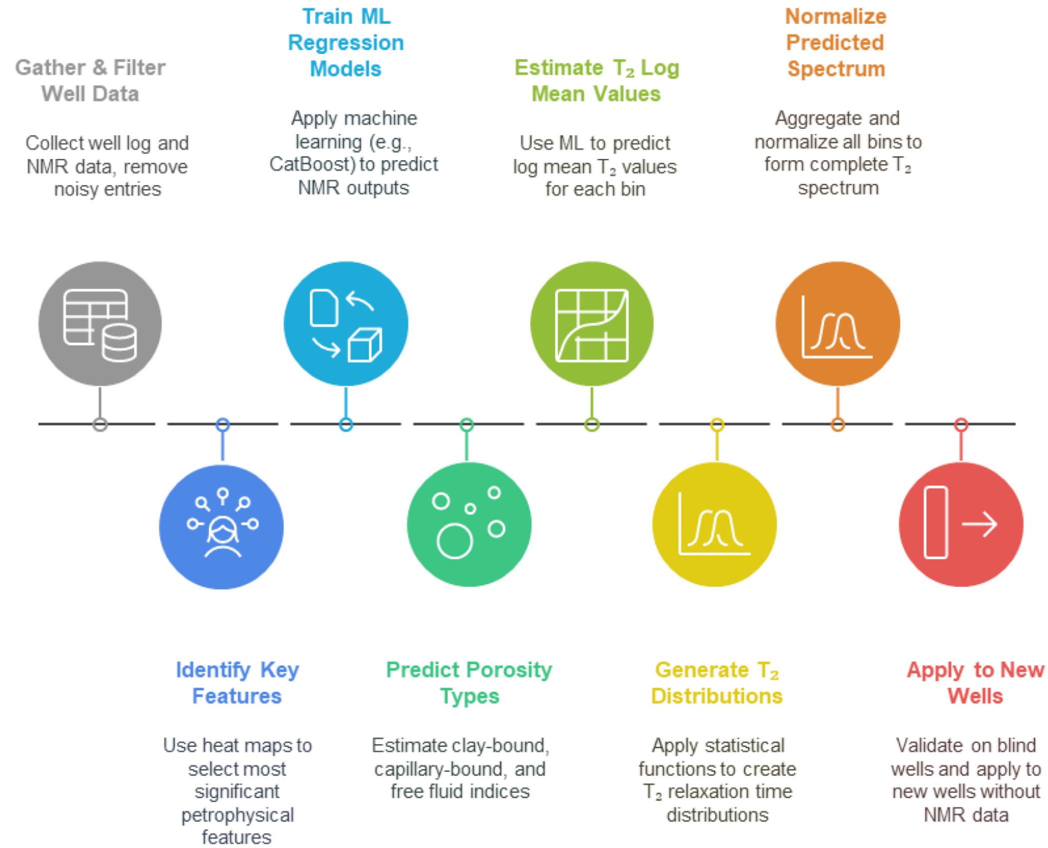

Figure 2—Workflow illustrating the sequential steps involved in Data preparation, ML based NMR output prediction, and Statistical Modelling for generating NMR T_2 distribution for the characterization of studied carbonate reservoirs.

Machine Learning (ML) Models

Machine learning (ML) algorithms enable computers to perform designated tasks by identifying underlying patterns from input data. In this context, python-based ensemble machine learning techniques were implemented to predict NMR log outputs using conventional wireline log data, providing a robust and flexible approach for high-dimensional petrophysical datasets.

- **Random Forest (RF)** is a widely adopted supervised machine learning algorithm effective for both classification and regression tasks. As an ensemble method, it generates multiple decision trees through the use of bagging or bootstrap aggregating, combining the predictions of these trees to form a 'forest' (Smith et al. 2013; Schonlau and Zou, 2020). For regression tasks, RF outputs the mean prediction of all trees, while for classification, the output corresponds to the majority class label. Its main hyperparameters—number of trees, maximum depth, and minimum node samples—can be tailored, but RF is renowned for delivering stable, high-quality predictions with minimal parameter adjustment.
- **Adaptive Boosting (AdaBoost)** is another sophisticated ensemble technique used for both classification and regression. AdaBoost works by sequentially training an ensemble of weak learners, typically small decision trees, incrementally focusing on previously misclassified data to reduce error (Al-Mudhafar et al. 2022; Ding et al. 2022; Radwan et al. 2022). The final result derives from a weighted combination of all learner outputs. Key hyperparameters include the number of estimators and the algorithm's learning rate. AdaBoost is highly effective and resilient, often achieving significant accuracy improvements with little tuning.
- **Gradient Boosting (GB)** refines model accuracy by building an ensemble of trees sequentially, each new tree trained to correct errors from the previous model (Di et al., 2019; Dahiya et al.,

2021; Otchere et al., 2022). High predictive performance is achieved by meticulously optimizing hyperparameters such as learning rate and tree depth. GB is prized for its precision and adaptability to various data types and structures.

- **Extreme Gradient Boosting (XGBoost)** further advances boosting techniques by incorporating efficient handling of sparse data, regularization, and parallel processing. XGBoost incrementally constructs decision trees to minimize a specified loss function, yielding robust results in complex regression and classification tasks (Gu et al. 2021; Bailey et al. 2021). Its main hyperparameters include tree depth, learning rate, and the number of estimators, with exceptional predictive power attainable even with basic settings.
- **Categorical Boosting (CatBoost)** specializes in processing categorical features efficiently, automating their encoding while building decision tree ensembles using symmetric tree structures and ordered boosting (Ding et al. 2021; Niu et al. 2021; Lu et al. 2022). CatBoost outperforms other models when dealing with datasets containing significant categorical variables, with major hyperparameters being the learning rate, tree depth, and iteration count.
- **Hyperparameter optimization** was used for optimizing parameters governing algorithm behaviors such as learning rate, tree depth, and estimator counter prior to training and crucial for maximizing generalization. The optimal hyperparameter configurations for all models were determined using the 'GridSearchCV' approach, automating extensive cross-validated searches (Gharavi et al., 2022). Table 1 details the chosen hyperparameter ranges and optimization process, ensuring robust, unbiased performance comparisons across models.

Table 1—Optimized Hyperparameter selection for each machine learning algorithm.

Algorithm	Hyperparameter Type	CBW	BVI	FFV	Kc	Ksdr	T_{2LM}
Random Forest	n_estimator	500	1000	500	1000	1000	1000
	Max depth	12	12	12	12	12	12
	others	*RF default					
GradientBoosting	n_estimator	100	100	100	100	100	100
	Learning rate	0.1	0.05	0.1	0.05	0.05	0.05
	Max depth	12	12	12	12	12	12
	others	*GB default					
CatBoost	Iterations	1000	1000	1000	1000	1000	1000
	Learning rate	0.05	0.01	0.05	0.03	0.03	0.05
	depth	9	15	9	13	13	13
	others	*CatBoost default					
AdaBoost	n_estimator	500	1000	1000	500	1000	1000
	Learning rate	0.1	0.12	0.1	0.15	0.1	0.1
	Max depth	10	10	10	10	10	10
	others	*AdaBoost default					
xGBoost	n_estimator	500	500	500	500	500	500
	Learning rate	0.2	0.05	0.2	0.1	0.1	0.15
	Max depth	13	15	13	15	15	15
	others	*xGBoost default					

Coefficient of Determination and Error Analysis

The performance of the ML models is assessed by examining their coefficient of determination (R^2), mean square error (MSE) and root-mean square error (RMSE) values at various stages, including training, testing, and blind data evaluation. The R^2 is defined as follows:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}\left(y_i - \hat{y}_i\right)^2}{\sum_{i=1}^{n}\left(y_i - \bar{y}\right)^2} \tag{9}$$

In this formula: y_i represents the observed (actual) value for data point i, $\hat{y}_i$ represents the predicted value for data point i, $\bar{y}$ is the mean of the observed values and n is the number of data points. The R^2 value provides insight into how well the predicted values match the actual values, with a perfect fit indicated by value of 1. MSE is the average of the square of the differences between the predicted and actual data.

$$MSE = \frac{1}{n}\sum_{i=1}^{n}\left(y_i - \hat{y}_i\right)^2 \tag{10}$$

MSE is consistently greater than or equal to 0. An MSE of 0 signifies perfect accuracy in predicting permeability, a practically unattainable ideal. Lastly, RMSE is defined as:

$$RMSE = \sqrt{MSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(y_i - \hat{y}_i\right)^2} \tag{11}$$

Table 2 presents a comparative performance analysis of machine learning models (RF, AdaBoost, Gradient Boosting, XGBoost, and CatBoost) based on R^2, MSE, and RMSE metrics for blind well datasets in predicting CBW, BVI, FFV, Kc, Ksdr, and T_{2LM}.

Table 2—Comparative Analysis of Machine Learning Models (RF, AdaBoost, Gradient Boosting, XGBoost, and CatBoost) Based on R^2, MSE and RMSE for Blind Well Datasets Predicting CBW, BVI, FFV, Kc, Ksdr, and T_{2LM}.

ML Model		Random Forest	GradientBoosting	CatBoost	AdaBoost	xGBoost
R^2	CBW	0.91	0.89	0.93	0.92	0.89
	BVI	0.66	0.66	0.73	0.63	0.64
	FFV	0.86	0.86	0.91	0.87	0.85
	Kc	0.83	0.84	0.88	0.85	0.84
	Ksdr	0.83	0.84	0.87	0.85	0.84
	T_{2LM}	0.75	0.75	0.84	0.78	0.76
MSE	CBW	0.00002	0.00003	0.00002	0.00002	0.00003
	BVI	0.00023	0.00022	0.00021	0.00024	0.00024
	FFV	0.00027	0.00028	0.00022	0.00025	0.00029
	Kc	0.37258	0.34674	0.30296	0.34243	0.35309
	Ksdr	0.33212	0.30966	0.26998	0.30741	0.31443
	T_{2LM}	0.01961	0.01938	0.01630	0.01718	0.01843
RMSE	CBW	0.015	0.014	0.014	0.015	0.015
	BVI	0.004	0.005	0.004	0.004	0.005
	FFV	0.016	0.016	0.014	0.015	0.017
	Kc	0.610	0.588	0.512	0.585	0.594
	Ksdr	0.576	0.556	0.519	0.554	0.560

ML Model		Random Forest	GradientBoosting	CatBoost	AdaBoost	xGBoost
	T_{2LM}	0.140	0.139	0.112	0.131	0.135

Statistical Modelling of NMR T_2 Spectra

Visual inspection of the pore size distribution on a semi-logarithmic plot revealed that the log of pore size appears to be approximately normally distributed in several carbonate porosity studies (Ausbrooks et al., 1999; Parra, Hackert, and Wilson, 2002). For spherical pores, NMR T_2 relaxation time is linearly related to pore size. This implies that T_2 relaxation times will follow a log-normal distribution like pore diameters (Kenyon, 1997; Coates et al., 1999). Carbonates have several pore shapes, but each normal component within our mathematical model will ideally represent one pore type in the sample, thus we only need the assumption of identical morphologies within one pore category. Based on prior studies that fitted Gaussian distribution to compute NMR outputs from core NMR data. In our study, we attempted to apply a reverse methodology by adding three Gaussian distributions (representing CBW, BVI, and FFV) to construct the NMR T_2 curve can provide a cost-effective and efficient means of determining the geological properties. For NMR spectroscopy, for a multimodal lognormal distribution, the signal amplitude can be expressed as (Genty et al., 2007; Ge et al., 2014),

$$S(t) = \sum_{i=1}^{n} \omega_i \frac{1}{\sqrt{2\pi \left(\log\sigma_i\right)^2}} e^{-\frac{\left(\log T_{2\,i} - T_{2mi}\right)^2}{2\left(\log\sigma_i\right)^2}} \tag{12}$$

Where, T_{2i} represents the T_2 time (ms); ω_i is the porosity volume of different pore classes. T_{2mi} is the log mean of each porosity class; σ_i is the standard deviation of the porosity components in the lognormal distribution model. Our python-based workflow and steps are presented as (Fig. 2):

a. Generate/predict the fluid volumes for clay-bound, capillary-bound, and free fluids, as well as their respective T_{2LM} values, utilizing the most effective machine learning-based method identified through comprehensive comparison.
b. Generate the Gaussian distribution corresponding to each porosity type.
c. Summation of results obtained for each fluid component to model the complete T_2 distribution.
d. Use the numerical integration and ensure the distribution is normalized.

Results

Porosity Prediction

Porosity classification from NMR T_2 distributions uses defined threshold values to differentiate Clay-Bound Water (CBW), Bulk Volume Irreducible (BVI), and Free Fluid Volume (FFV). CBW, water adsorbed on clay surfaces, has T_2 values between 0.3 ms and 3 ms. Key predictors for CBW include sonic transit time (DTCO), shale volume (V_{sh}), neutron porosity (NPHI), and density-neutron porosity difference (DPHI) (Fig. 3). CatBoost outperforms other models like AdaBoost and Random Forest in accuracy across training, testing, and validation (Figs. 4 and 5). BVI, representing immobile water trapped by capillary forces with T_2 times from 3 to 100 ms in carbonates, is best predicted by CatBoost. Important inputs include DTCO, DPHI, NPHI, and bulk density (RHOB). FFV corresponds to fluids in larger pores with T_2 times above 100 ms up to 3000 ms. CatBoost again shows highest accuracy, influenced mainly by effective porosity (PHIE), RHOB, NPHI, resistivity (RT), DPHI, and Vsh. Unlike CBW and FFV, BVI shows weaker correlations with logs, making its prediction more challenging in carbonates and sandstones, as noted in prior studies.

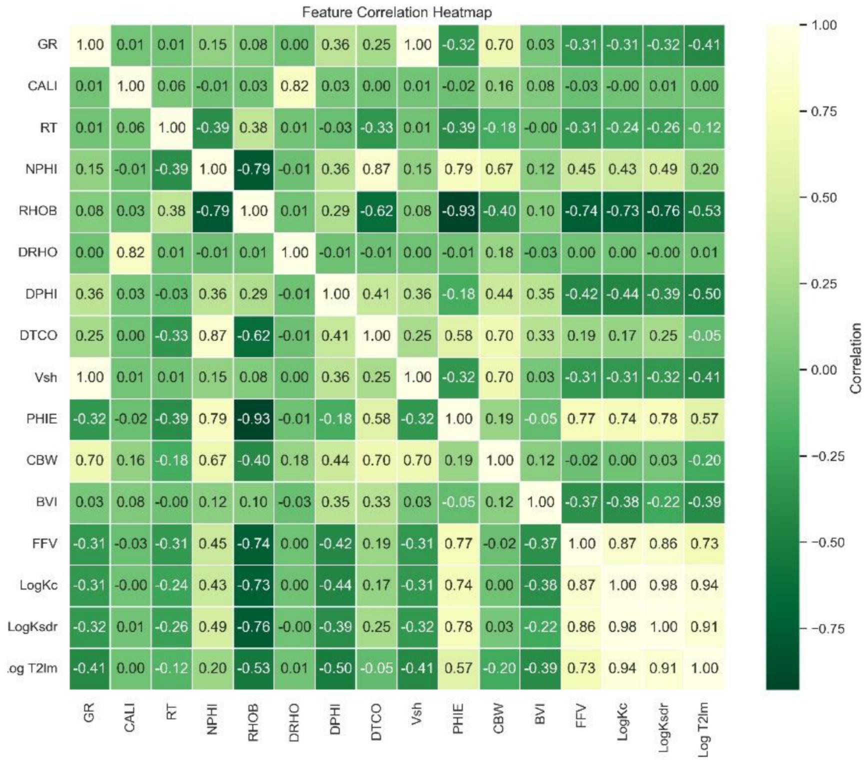

Figure 3—Cross-correlation matrix displaying feature selection. Light colors indicate positive correlations, while dark colors indicate negative correlations. Refer to the Web version of this article for color interpretation.

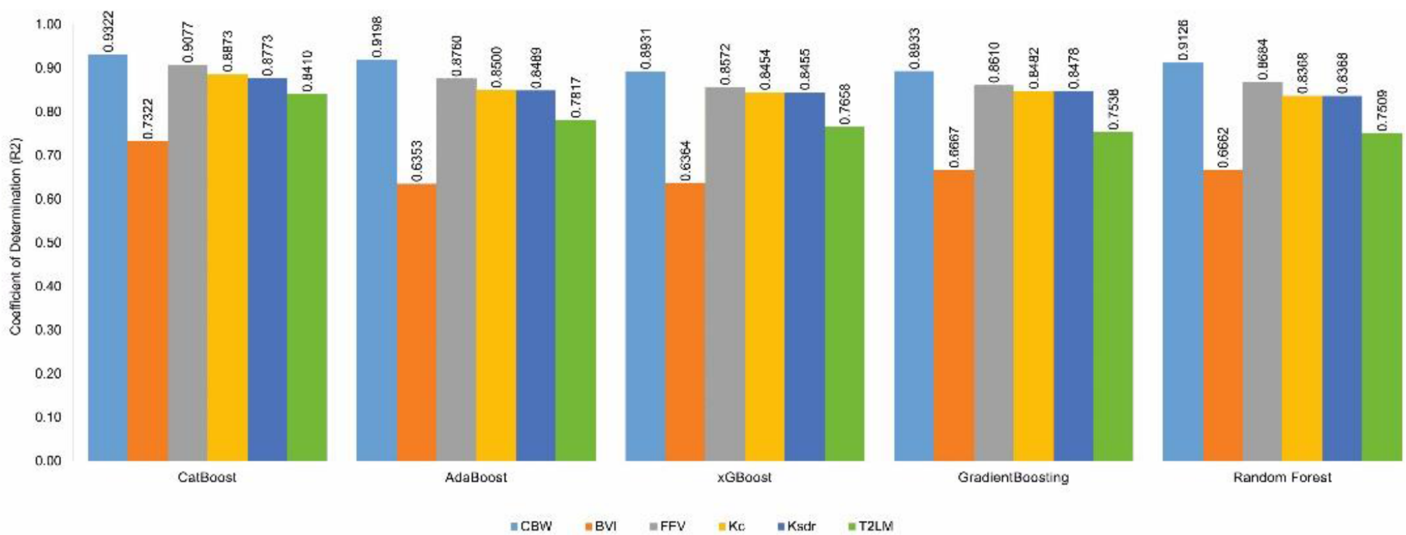

Figure 4—Comparative analysis of the performance of various machine learning models, including Random Forest (RF), AdaBoost, Gradient Boosting, XGBoost, and CatBoost. The evaluation is based on R^2 scores across blind well datasets, providing insights into the prediction accuracy for CBW, BVI, FFV, Kc, Ksdr and T_{2LM}.

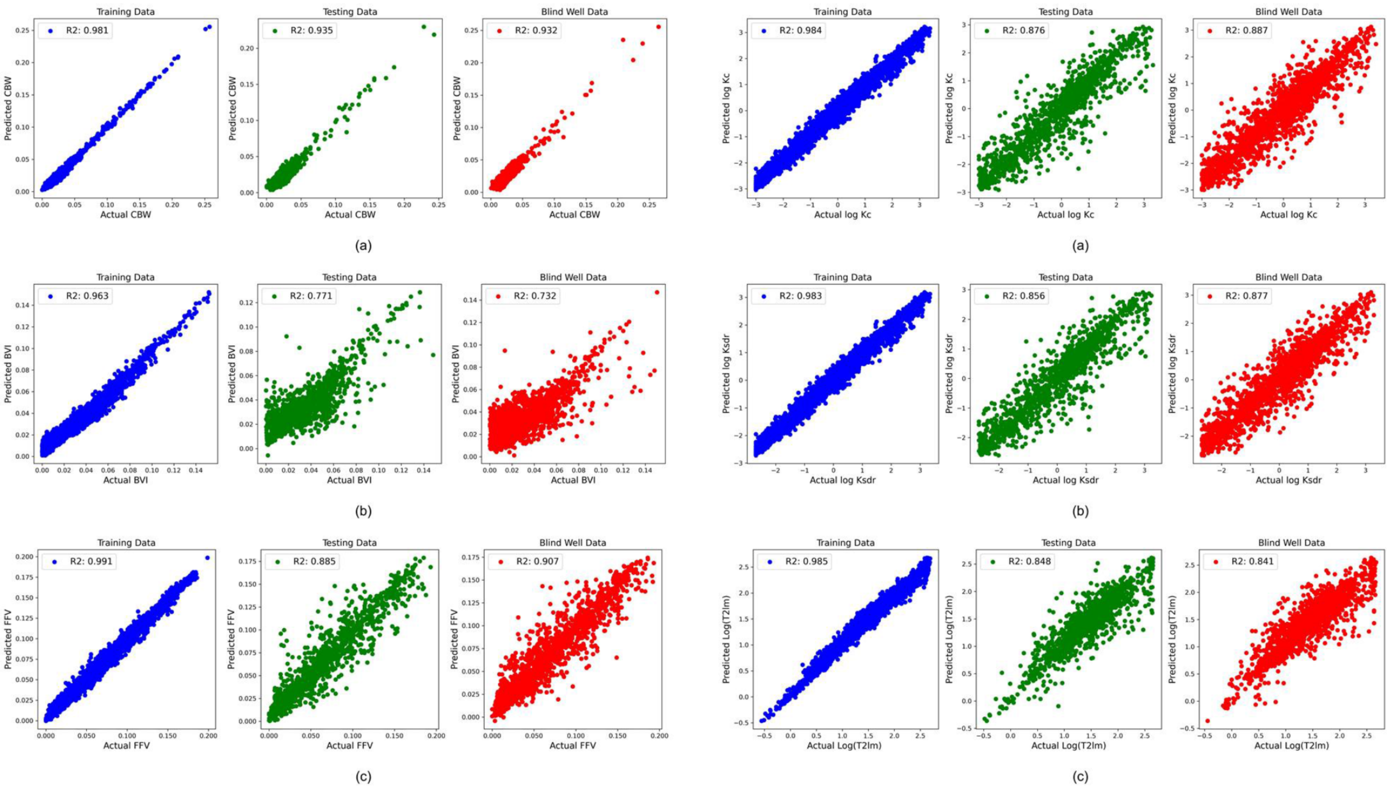

Figure 5—Cross-plot comparison of predicted versus actual values for CBW, BVI, FFV, Kc, Ksdr, and T_{2LM} using CatBoost across training, testing, and blind validation datasets.

Permeability Prediction

This study targets prediction of industry-standard NMR permeability metrics, specifically Coates permeability (K_c) and SDR permeability (K_{SDR}). Results depicted in Figs. 4 and 5 demonstrate that CatBoost outperforms competing models—including AdaBoost, Gradient Boosting, and XGBoost—especially during blind validation. Feature correlation analysis (Fig. 3) identifies PHIE, RHOB, NPHI, and DPHI as the dominant controlling parameters for permeability prediction.

Prediction of NMR T_2 Logarithmic Mean (T_{2LM})

The logarithmic mean of T_2 relaxation time, T_{2LM}, serves as a key indicator of pore size distribution in reservoir rocks. Machine learning models were employed to predict T_{2LM}, with CatBoost achieving the highest accuracy on both training and blind validation datasets, followed by AdaBoost and XGBoost (see Figs. 4 and 5). Correlation analysis identified effective porosity (PHIE), bulk density (RHOB), and the density-neutron porosity difference (DPHI) as the most influential input features in predicting T_{2LM}.

According to Equation 12, T_{2LM} is calculated as the logarithmic mean of the T_2 relaxation times associated with distinct porosity components: Clay-Bound Water (CBW), Bulk Volume Irreducible (BVI), and Free Fluid Volume (FFV). Specifically, the equation aggregates the mean T_2 values of each class—referred to as T_{2mi}—weighted by their corresponding porosity fractions.

To accurately capture the pore structure, class-specific T_{2mi} values were derived from the training dataset. These values reflect the average T_2 behavior within each porosity class:

- T_{2LMCBW} (mean T_2 of CBW) typically exhibits a unimodal distribution in the 0.3–3 ms range, and was treated as a constant in this study, with an average value of 1.13 ms.

- T_{2LMBVI} (mean T_2 of BVI) shows a bimodal distribution in highly porous zones due to mixed capillary-bound pore systems, ranging from 3–100 ms. Because of this complexity, a single average value is difficult to define.
- T_{2LMFFV} (mean T_2 of FFV) also presents a bimodal character in porous zones but demonstrates a strong correlation with total porosity in water-wet intervals (see Fig. 6).

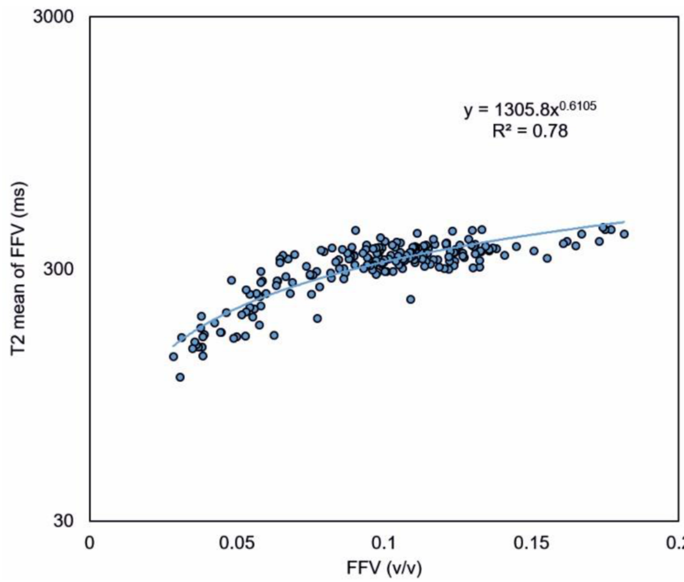

Figure 6—The relationship between FFV and corresponding T_2 mean value (T_{2LMFFV}: computed within 100-3000ms) demonstrates a strong association, as indicated by the regression line, suggesting a robust power law relationship.

While the overall T_{2LM} is predicted directly via machine learning, the approach aims to reconstruct the underlying T_2 distribution by estimating individual component means:

- Use a fixed value for T_{2LMCBW}.
- Predict T_{2LMFFV} using its empirical correlation with porosity.
- Calculate T_{2LMBVI} using Equation 12 by back-solving, given the known values of T_{2LM}, CBW, and FFV components.

This decomposition enables a more nuanced interpretation of the pore system, allowing the reconstructed T_{2LM} to reflect the contributions from different fluid-bound pore types.

Discussion

Our analysis reveals that the CatBoost machine learning model consistently outperforms other algorithms in predicting key reservoir properties, including Clay-Bound Water porosity (CBW), Free Fluid Volume porosity (FFV), permeability (Coates and SDR methods), and the logarithmic mean of the T_2 distribution (T_{2LM}) across blind datasets. The model achieves high predictive accuracy with R^2 values predominantly above 0.85 (Figs. 4 and 5). In contrast, prediction accuracy for Bulk Volume Irreducible (BVI) porosity is comparatively lower, with an R^2 near 0.73. This discrepancy is attributable to BVI's weaker correlation with conventional well log inputs such as shale volume (Vsh), neutron porosity (NPHI), density-neutron porosity difference (DPHI), and effective porosity (PHIE), which show stronger relationships with CBW and FFV. The lower predictability of BVI aligns with findings from previous studies on both carbonate and sandstone reservoirs, underscoring the inherent challenges in modeling this property. AdaBoost and XGBoost models consistently rank below CatBoost in performance for both training and blind datasets

(Figs. 4 and 5), highlighting the superior robustness of CatBoost's gradient boosting framework in this domain.

Fig. 7 demonstrates CatBoost's predictive capability through a comparison between measured and predicted NMR outputs (marked as red dots) for a blind interval of Well 1. The predicted CBW aligns closely with observed values, while FFV, T_{2LM}, and permeability predictions also exhibit excellent agreement. Minor discrepancies occur in BVI predictions, especially in intervals with elevated original BVI values. Overall, the predicted NMR outputs reliably capture the responses across varied pore fluid types and shale contents. Building on these predictions, synthetic field-equivalent NMR T_2 distributions were generated by combining predicted CBW, BVI, FFV volumes and their respective mean T_2 values. Fig. 8 compares the modeled and field-recorded NMR spectra within the same reservoir intervals, showing close agreement in peak positions and distribution shapes. The model is effective in representing both clay-rich and clay-poor intervals and reproduces characteristic bimodal and quasi-bimodal lognormal distribution corresponding to capillary-bound and free fluid pore volumes. The use of 3 ms and 100 ms thresholds for capillary-bound and free fluid water in carbonates remains valid and is preserved in the synthetic data, which validates the robustness and practical applicability of our ML-driven approach.

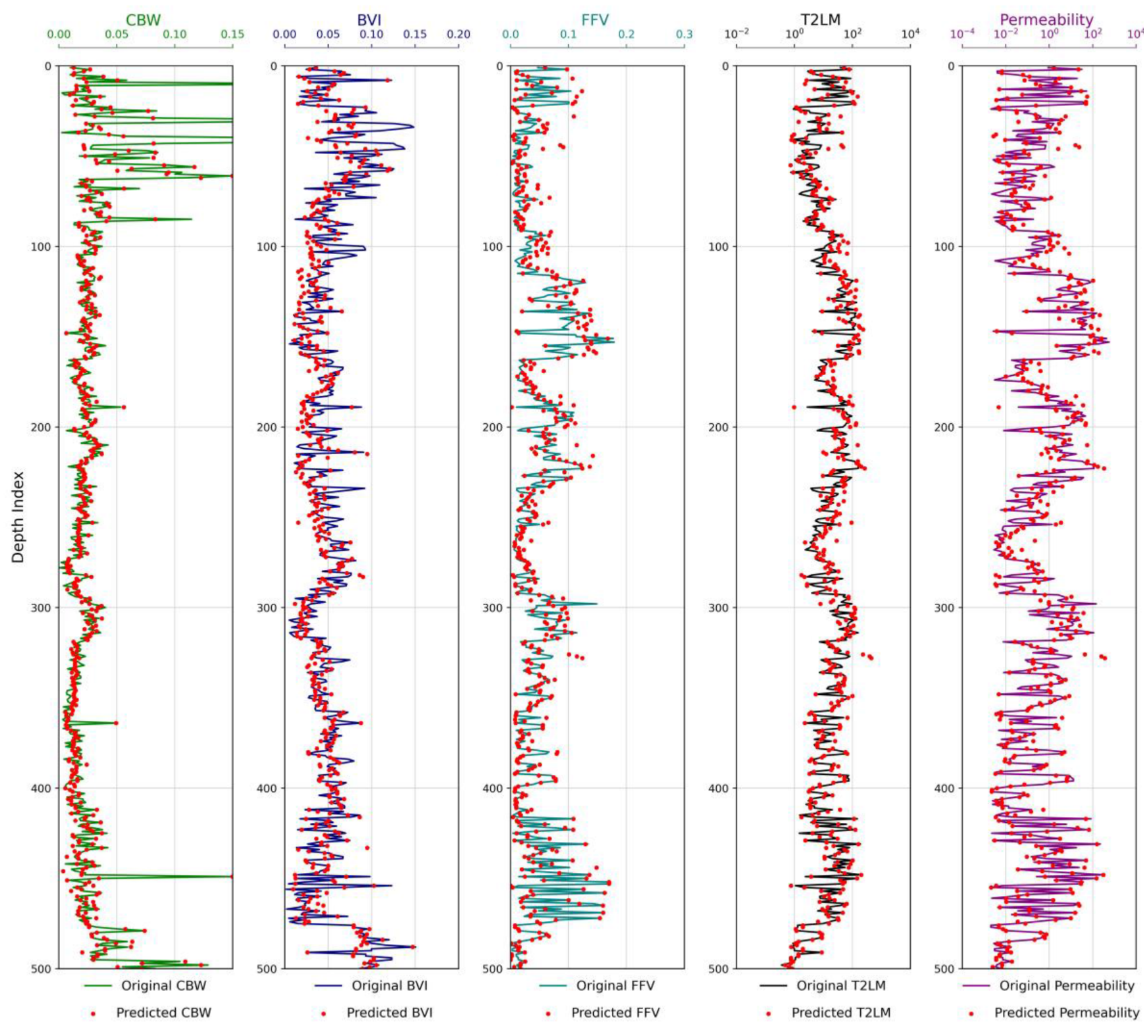

Figure 7—Comparison of real and predicted NMR outputs (red dots) in a blind well interval (Well 1). CBW, BVI, FFV, T_{2LM}, and permeability (Kc) tracks are illustrated. The predicted CBW closely aligns with actual values, while FFV, T_{2LM}, and permeability exhibit commendable agreement. Discrepancies are noted in predicted BVI data, particularly with slightly elevated original BVI porosity.

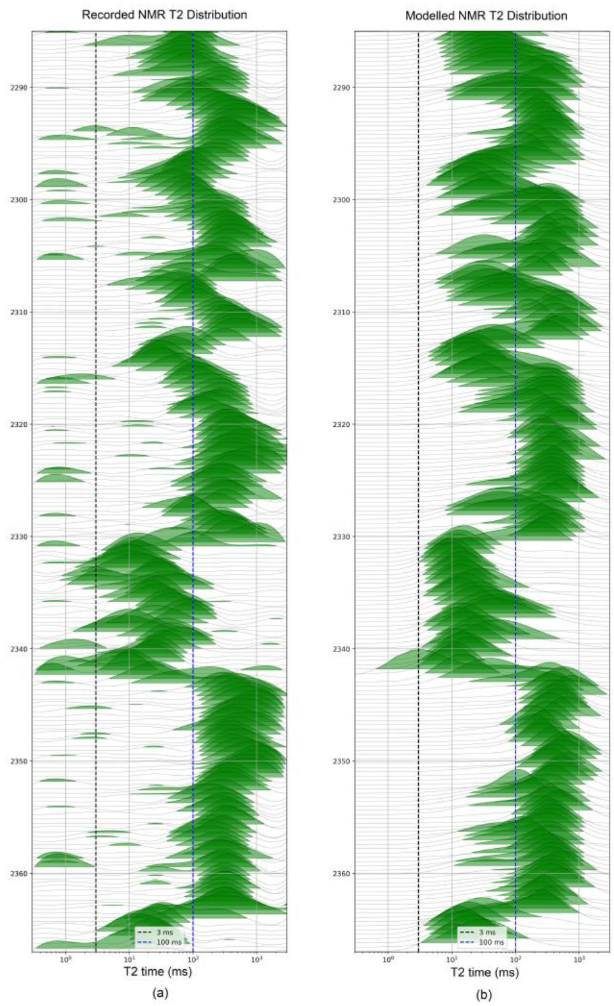

Figure 8—Comparison of field-recorded and modelled NMR T_2 distributions in the same carbonate reservoir interval, highlighting similarities in peak positions and overall patterns, including representations of capillary pore volume and free fluid porosity, demonstrating the reliability of the current innovative workflow.

In the study region, majority of wells lack NMR logs due to limited commercial availability before 1995 but have conventional logs such as gamma ray, resistivity, neutron, density, and sonic. Leveraging these, we predicted NMR outputs and constructed NMR T_2 distributions. Fig. 9 displays wireline log responses alongside predicted NMR outputs and modeled T_2 distributions for Well 2. Most intervals indicate minimal shale except for a zone near 2,200 m depth. Effective porosity averages around 10%, with localized drops below 3% in tight limestone layers interspersed within more porous formations. A detailed segment of the T_2 distribution (Fig. 9b) reveals a nearly constant clay-bound porosity alongside variable BVI and FFV. Increased FFV coincides with decreased BVI, producing a unimodal T_2 distribution extending beyond 100 ms. Conversely, when FFV declines, BVI either rises or remains stable, resulting in a bimodal T_2 pattern characterized by peaks between 3–100 ms and above 100 ms. Further reductions in FFV lead to a unimodal T_2 distribution predominantly within 3–100 ms, reflecting a capillary-bound water phase with elevated irreducible water saturation.

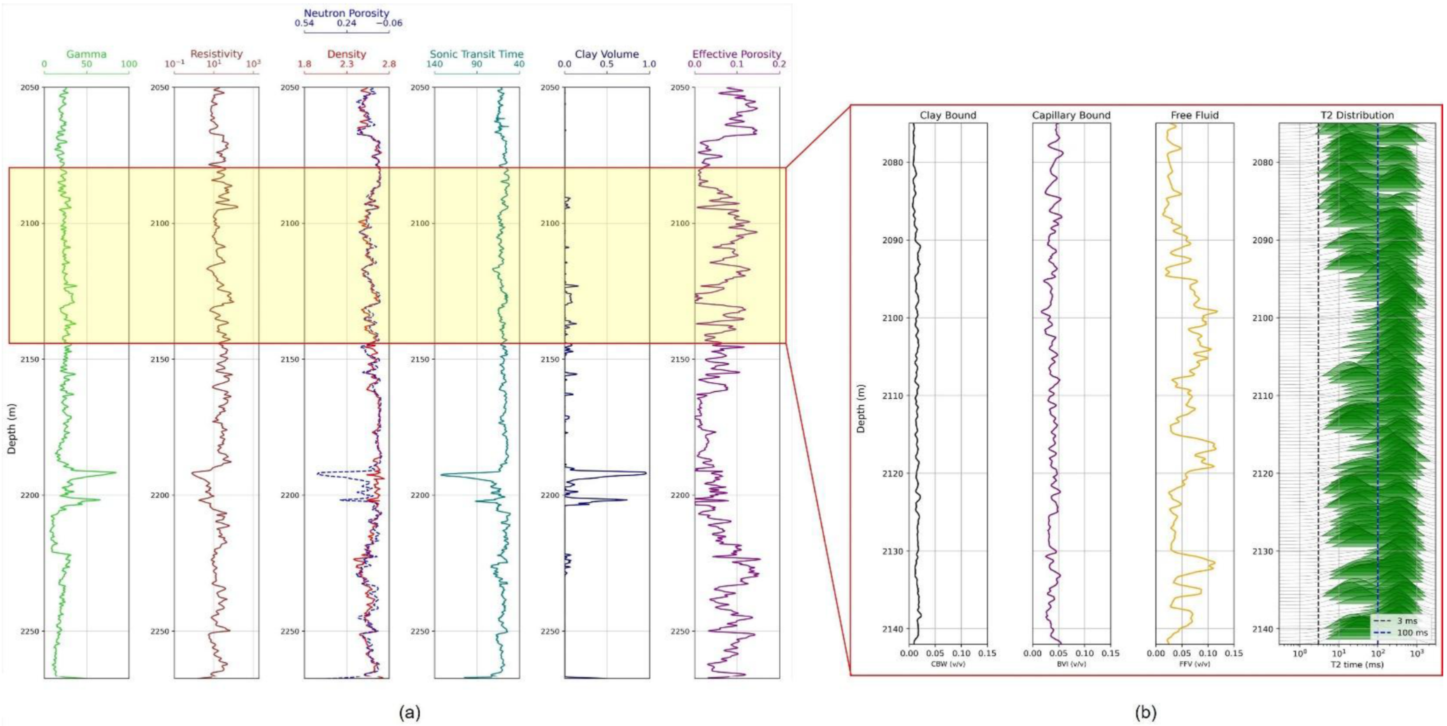

Figure 9—Wireline log response of Well 2, new workflow based predicted NMR outputs, and modelled NMR T_2 distribution, highlighting intervals shows variations in Capillary-bound (BVI) and free fluid (FFV) porosity, showcasing unimodal and bimodal T_2 distribution patterns.

This comprehensive evaluation confirms that our ensemble machine learning workflow, particularly CatBoost, effectively predicts NMR properties and reconstructs T_2 distributions, offering a valuable tool for enhanced reservoir characterization where direct NMR logging is unavailable.

Conclusions

- NMR logs offer comprehensive data regarding the distribution of pore volume, permeability, and irreducible water saturation.
- The NMR log or NMR data is not available due to the unavailability of technology in old wells, significant expenses involved, and its inapplicability to numerous scenarios.
- Machine learning algorithms such as Random Forest, AdaBoost, Gradient Boosting, XGBoost, and CatBoost have been employed to create models for standard NMR output and T_2 spectrum.
- CatBoost surpasses all other methods, resulting in the most accurate prediction of standard NMR outputs with lowest error.
- First time so far, statistical modelling has been used in generating complete T_2 distribution using predicted NMR outputs. Both observed and modelled T_2 spectra exhibit strong similarities, especially in single-peaked distribution in terms of peak placements and overall pattern across the spectrum.
- The proposed workflow also enhances conventional log interpretation by partitioning porosity into shale-bound, capillary-bound, and free-fluid fractions, improving permeability estimates and enabling synthetic capillary pressure curves when calibrated with core data.
- Despite these improvements, prediction accuracy decreases in intervals with extreme BVI values due to limited representation in the training dataset. The models are also sensitive to input data quality and may require recalibration for wells with significantly different lithological or fluid properties.

- Water saturation estimation was not included, as the current model focuses on pore characterization. This is likely explaining why longer T_2 amplitudes are under-resolved in some intervals due to possible fluid effects. Future work may incorporate bulk and diffusion relaxation to improve T_2 modeling.

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